23MCAJ301	Machine Learning	L	T	P	J	S	C	Year of Introduction
		2	0	2	2	5	5	2023

Preamble: This is an introductory course on basic concepts behind various machine learning techniques. Machine learning is the study of adaptive computational systems that improve their performance with experience. At the end of the course the students should be able to design and implement machine learning solutions to classification, regression, and clustering problems and to evaluate and interpret the efficiency of the algorithms.

Prerequisite: Python programming

Course Outcomes: After the completion of the course the student will be able to

CO1	Understand the basics of machine learning and data visualization techniques.
CO2	Apply the principle behind lazy learning and probabilistic learning algorithms to solve real life problems.
CO3	Apply the principle behind decision trees & regression methods to solve data science problems.
CO4	Solve real life problems using neural networks and support vector machines.
CO5	Analyse the performance of various machine learning models.

CO - PO MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3			3							
CO2	3	3	2		3	3	3	3	3		3	
CO3	3	3	2	2	3	3	3	3	3		3	
CO4	3	3	3	2	3	3	3	3	3		3	
CO5	3	3	3	2	3	3	3	3	3		3	

Assessment Pattern for Theory component

Bloom's Category	Continuous Assessment Tools			End Semester Examination
	Test1	Test2	Other tools	
Remember		✓	✓	✓
Understand		✓	✓	✓
Apply		✓	✓	✓
Analyse			✓	
Evaluate			✓	
Create			✓	

Assessment Pattern for Lab component

Bloom's Category	Continuous Assessment Tools	
	Class work	Test1
Remember		
Understand	✓	✓
Apply	✓	✓
Analyse	✓	✓
Evaluate		
Create		

Assessment Pattern for Project Component			
Bloom's Category	**Continuous Assessment Tools**		
	Evaluation1	**Evaluation2**	**Report**
Remember			
Understand	✓	✓	
Apply	✓	✓	
Analyse	✓	✓	
Evaluate	✓	✓	
Create		✓	

Mark Distribution of CIA

Course Structure [L-T-P-J]	Attendance	Theory [L- T]		Practical [P]	Project [J]			Total Marks
		Assignment	**Test-2**	**Class work**	**Evaluation 1**	**Evalaution-2**	**Report**	
2-0-2-2	5	10	15	10	5	10	5	**60**

Total Marks distribution

Total Marks	CIA (Marks)	ESE (Marks)	ESE Duration
100	60	40	2.5 hours

SYLLABUS

MODULE I: Introduction to Machine learning (5hrs)

Introduction to Machine Learning - How do machines learn – Understanding data:- mean, median, mode, Measuring spread. Review of distributions: Uniform and normal. Data Visualization- Histogram, Box plot, Scatter plot. Evaluating model performance: Confusion matrices, Precision and recall, Sensitivity and specificity, F-measure, ROC curves, Cross validation - K-fold cross validation, Bootstrap sampling.

MODULE II : Lazy Learning and Probabilistic Learning (5hrs)

Lazy learning: Classification using K-Nearest Neighbour algorithm - Measuring similarity with distance, Choice of k, Preparing data for use with k-NN.
Probabilistic learning: Understanding Naive Bayes - Conditional probability and Bayes theorem, Naive Bayes algorithm for classification, The Laplace estimator, Using numeric features with Naive Bayes.

MODULE III : Decision trees and Regression (5hrs)

Classification Using Decision Trees and Rules- Divide and conquer strategy. Decision tree algorithm.
Regression Methods - Simple linear regression - Ordinary least squares estimation Correlations - Multiple linear regression.

MODULE IV : Neural Networks and SVM (5hrs)

Neural network learning: Artificial neurons, Activation functions, Network topology, Training neural networks with back propagation.
Support vector machines: Hyperplanes, Classification using hyperplanes, Maximum margin hyperplanes in linearly separable data, Using kernels for non-linear spaces.

MODULE V : Clustering and Model Evaluation (4hrs)
Clustering: The k-means clustering algorithm, Using distance to assign and update clusters, Choosing number of clusters. Evaluate the performance of different machine learning algorithms. Improving model performance - Bagging, Boosting, Random forests.

Text books

1. Brett Lantz, Machine Learning with R, Second edition, PackT publishing 2015
2. Vijay Kotu, Bala Deshpande, Data Science Concepts and Practice, Morgan Kaufmann Publishers 2018
3. Machine Learning- Course Materials http://cs229.stanford.edu/materials.html

Suggested MOOC

1. https://www.coursera.org/learn/machine-learning
2. https://onlinecourses.nptel.ac.in/noc23_cs87/preview

Reference books

1. Michael Steinbach, Pang-Ning Tan, and Vipin Kumar, Introduction to Data Mining, Pearson 2016.
2. Jiawei Han, Micheline Kamber and Jian Pei, Data mining Concepts and techniques, Morgan Kaufmann Publishers 2012.
3. Peter Harrington, Machine Learning in action, Dreamtech publishers 2012.
4. Dr M Gopal, Applied Machine learning, McGraw Hill Education Private Limited
5. E. Alpayidin, Introduction to Machine Learning, Prentice Hall of India (2005)
6. T. Hastie, RT Ibrashiran and J. Friedman, The Elements of Statistical Learning, Springer 2001.
7. Data Science from Scratch: First Principles with Python, Joel Grus, O'Reilly, First edition, 2015.
8. Introduction Data Science, Davy Cielen, Arno D. B. Meysman, Mohamed Ali, Manning Publications Co., 1st edition, 2016.

COURSE CONTENTS AND LECTURE SCHEDULE

No.		No. of Hours
	MODULE 1	
1.1	Basics of Machine learning-How do Machine learn	1
1.2	Understanding data:- numeric variables – mean, median, mode, categorical variables. Measuring spread. Review of distributions: Uniform and normal.	1
1.3	Data Visualization- Histogram, Box plot, Scatter plot.	1
1.4	Evaluating model performance: Confusion matrices, Precision and recall, Sensitivity and specificity.	1
1.5	F-measure, ROC curves, Cross validation - K-fold cross validation, Bootstrap sampling.	1
	MODULE II	
2.1	Lazy learning: Classification using K-Nearest Neighbor algorithm, Measuring similarity with distance	1
2.2	Choice of k, Preparing data for use with k-NN.	1
2.3	Probabilistic learning: Understanding Naive Bayes - Conditional	1

	probability and Bayes theorem	
2.4	Naive Bayes algorithm for classification	1
2.5	The Laplace estimator, Using numeric features with Naive Bayes.	1
MODULE III		
3.1	Classification Using Decision Trees and Rules- Divide and conquer strategy.	1
3.2	Decision tree algorithm.	1
3.3	Decision tree algorithm –Solving examples	1
3.4	Regression Methods - Simple linear regression - Ordinary least squares estimation	1
3.5	Solving problems- Correlations - Multiple linear regression	1
MODULE IV		
4.1	Neural network learning- Artificial neurons, Activation functions, Network topology	1
4.2	Training neural networks with back propagation.	1
4.3	Solving problems	1
4.4	Support vector machines-Hyperplanes, Classification using hyperplanes, Maximum margin hyperplanes in linearly separable data.	1
4.5	Using kernels for non-linear spaces	1
MODULE V		
5.1	Clustering: The k-means clustering algorithm, Using distance to assign and update clusters, Choosing number of clusters.	1
5.2	Solving examples	1
5.3	Evaluate the performance of different machine learning algorithms.	1
5.4	Improving model performance - Bagging, Boosting, Random forests.	1

LESSON PLAN FOR LAB COMPONENT

No.	Topic	No. of Hours	Experiment
1	Study of Machine Learning libraries in python	2	Implement programs listed under CO1
2	Data Summarization and Visualization	2	
3	k-NN	2	Implement programs listed under CO2
4	Naive Bayes	2	
5	Decision Tree	2	Implement programs listed under CO3
6	Linear Regression	2	
7	Multiple Regression	2	
8	Feed forward Neural Network	2	Implement programs listed under CO4
9	SVM	2	
10	K means	2	Implement programs listed under CO5
11	Performance Evaluation and improvement -Random forest	2	

12	Compare different classification algorithms on real world dataset	2	

List of Experiments

CO1

1. Programs to handle data using NumPy, Pandas and matplotlib/ plotly/ bokeh/ seaborn for data visualization.
2. Calculate the mean, median, and mode of a given numeric dataset.
3. Plot a histogram and a boxplot for any dataset to visualize the data distribution.
4. Explore a real-world dataset and summarize its key characteristics using descriptive statistics and data visualization.

CO2

1. You are given a dataset that contains missing values in some of its features. Your task is to explore the dataset, identify the missing values, and discuss possible strategies to handle them when using the k-Nearest Neighbour (k-NN) algorithm for classification.(Use any publicly available dataset, plot the decision boundary for different k values and observe the impact on the model's performance.)
2. You are given a dataset with a mix of categorical and continuous numeric features. Your task is to implement the Naive Bayes algorithm for a binary classification problem, considering the presence of both types of features. Additionally, you will explore and apply different strategies to handle continuous numeric features in the Naive Bayes algorithm.(Explore different datasets to perform three variants of naive Bayes: Gaussian Naive Bayes, Multinomial Naive Bayes, and Bernoulli Naive Bayes.)

CO3

1. Program to implement decision trees using and standard dataset available in the public domain and find the accuracy of the algorithm.(Implement the pruning technique to avoid overfitting and re-evaluate the decision tree's performance after pruning. Compare the decision tree model's performance with other classification algorithms, such as k-Nearest Neighbors (k-NN) or Naive Bayes. Use either the ID3, C4.5, or CART (Gini impurity) algorithm)
2. Explore the concepts of simple linear regression, multiple linear regression, and correlations using the ordinary least squares estimation method to fit regression models.
3. Work with a dataset containing independent variables (features) and a dependent variable (target) to predict and analyze their relationships.(Implement feature scaling (e.g., standardization or normalization) for the independent variables and re-evaluate the performance of the multiple linear regression model. Implement regularization techniques (e.g., Lasso or Ridge regression) to handle potential overfitting in the multiple linear regression model. Compare the performance of regularized and non-regularized models.)

CO4

1. Programs on feed forward network to classify any standard dataset available in the public domain.
2. Programs on SVM to classify any standard dataset in the public domain.

CO5

1. Programs to implement k-means clustering technique using any standard dataset available in the public domain.

2. Programs to implement Random Forest technique using any standard dataset available in the public domain.
3. Compare the performance evaluation of different classification algorithms.

Sample project topics

1	Apply KNN, Naïve Bayes and Decision tree classifier on Breast cancer dataset and compare the performance using different metrics. Identify the optimum set of parameters for the algorithms using parameter tuning method and compare the performance.
2	Predict medical expenses of people based on insurance data set. Identify the correlations and techniques to boost the accuracy of the model.

LESSON PLAN FOR PROJECT COMPONENT

Total No. of Class Hours: 24 12 Hours of self-study hours also should be utilized for the development of the complete project.		
No.	**Topic**	**No. of Class Hours[24]**
1	Preliminary Design of the Project	4
2	Zeroth presentation (4th week)	2
3	Project work - First Phase	4
4	Interim Presentation (7th and 8th weeks)	4
5	Project work - Final Phase & Report writing (discussions in class during project hours)	6
6	Final Evaluation and Presentation (11th and 12th weeks)	4

CO Assessment Questions	
	CO1
1	Differentiate between supervised and unsupervised learning algorithms
2	Explain the typical methods to visualize data
3	Illustrate using a standard data set, how cleaning and selection of right features are done.
	CO2
4	Explain how to apply k-NN classifier in a data science problem.
5	Use Naive Bayes algorithm to determine whether a red domestic SUV car is a stolen car or not using the following data:

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

CO3

6 Explain how to estimate a linear regression model.

7 Consider the following set of training examples

Find the entropy of this collection of training examples with respect to the target function “classification”?

Calculate the information gain of **a2** relative to these training examples?

Instance	**Classification**	**a1**	**a2**
1	+	T	T
2	+	T	T
3	-	T	F
4	+	F	F
5	-	F	T
6	-	F	T

8 Explain how to simplify a decision tree by pruning

CO4

9 Explain how artificial neural networks mimic human brain to model arbitrary functions and how these can be applied to real-world problems.

10 Explain how a support vector machine can be used for classification of linearly separable data.

11 Explain how the kernel trick is used to construct classifiers in nonlinearly separated data.

CO5

12 Find the three clusters after one epoch for the following eight examples using the *k*- means algorithm and Euclidean distance: A1 = (2,10), A2 = (2,5), A3 = (8,4), A4 = (5,8), A5 = (7,5), A6 = (6,4), A7 = (1,2), A8 = (4,9). Suppose that the initial seeds (centres of each cluster) are A1, A4 and A7.

13 Suppose 10000 patients get tested for flu; out of them, 9000 are actually healthy and 1000 are actually sick. For the sick people, a test was positive for 620 and negative for 380. For the healthy people, the same test was positive for 180 and negative for 8820. Construct a confusion matrix for the data and compute the precision and recall for the data.